Department of Electrical and Electronics Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: EE3037 Power System Transients

Name of the Faculty member: Dr.D.Karthik Prabhu

Innovative Practice Description

- **Unit / Topic:** Unit I / Sources of different types of transients
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO1
- **Activity Chosen:** One Minute Paper

- **Justification:**

The chosen topic – sources of different types of transients in power system network. Since, in power system different type of transient will occur in different situation. After teaching the concept, I thought of conducting this activity for enabling the students to understand the different types of transients which enhance their learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 10 Minutes

- **Details of the Implementation:**

After completing the topic, I gave 5 minutes for the students to think about the topic without writing anything.

Total Strength: 53

Reporter : Myself

At the end of the class

- ✓ I asked the students to think about various types of transients for 5 minutes.
- ✓ I told them to write as much as they remember for 3 minutes.
- ✓ Finally, I collected the papers from the students.(2 minutes)

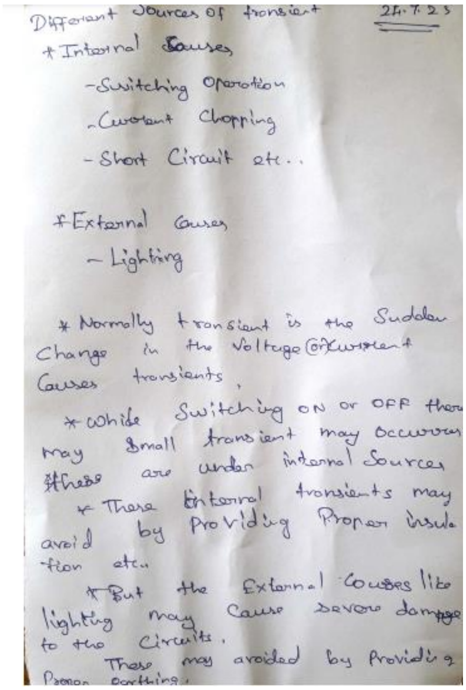
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO10	PO12	PSO1
CO1	3	2	2	3	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	**PO10**
	1
Justification for correlation	Due to this activity, student's written communication will be enhanced. So it is slightly correlated.

- **Screenshot of the activity:**


Different Sources of transient 24.7.23
* Internal Causes
-Switching Operation
-Current Chopping
-Short Circuit etc..
* External Causes
– Lighting
* Normally transient is the Sudden Change in the Voltage (or) current Causes transients.
* While Switching ON or OFF there may small transient may occurring these are under internal Sources
* These internal transients may avoid by providing Proper insulation etc..
* But the External Causes like lighting may Cause severe damage to the circuits.
These may avoided by providing Proper earthing.

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

Students felt that writing about the given topic make them understand it well and able to recollect immediately.

❖ ***Benefit of the practice:***

1. All the students were able to write the given topic.
2. After conducting the activity, I came to know that students understood the topic and they were able to explain it.

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting the activity
2. Some students could not be able to recollect and write immediately.

References:

✓ Allan Greenwood, 'Electrical Transients in Power Systems', Wiley Inter Science, New York, 2nd Edition, 1991.
✓ Pritindra Chowdhari, "Electromagnetic transients in Power System", John Wiley and Sons Inc.,Second Edition, 2009.
✓ https://oncourseworkshop.com/self-awareness/one-minute-paper/

Signature of Faculty Member **HOD**

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Course Code & Title: EE3037 Power System Transients

Name of the Faculty member: Dr.D.Karthik Prabhu

Innovative Practice Description

- **Unit / Topic:** Unit II / Normal and Abnormal transients
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO6
- **Activity Chosen:** Think Pair Share

- **Justification:**

The chosen topic – provides understanding about the normal transients and abnormal transients. After teaching the concept, I thought of conducting this activity for making the students to provide knowledge about the transients and its effects which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 10 Minutes

- **Details of the Implementation:**

After completing the topic, the students will be paired with their neighbors, 4 students as a group.

Total Strength is 54, Number of Pairs – 14

Photographer: Myself

Reporter: Myself

At the end the Class (Last 10 minutes)

- I asked the students to **think** about normal and abnormal transients, and its effects for 2 minutes.
- Then I told them to **Pair** with their neighbors and discuss about the concept of normal and abnormal transients for another 2 minutes.
- Finally, I selected 1 Pair from each column randomly and ask them to **share** about normal and abnormal transients. (6 minutes)
- Finally, I summarized the points again about normal and abnormal transients.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PO12	PSO1
CO2	3	3	3	3	1	1	1	3

- **PO / PSO mapped:**

<table>
<tr><td rowspan="2">Innovative practice</td><td>PO9</td></tr>
<tr><td>1</td></tr>
<tr><td>Justification for correlation</td><td>Due to this activity, student’s team work is getting enhanced. So it is slightly correlated.</td></tr>
</table>

- **Screenshot of the activity:**

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

Students felt that discussing with the team members and sharing the concept with everyone make them understand it well and enhanced their knowledge.

❖ ***Benefit of the practice:***

1. All the students were able to share the topic given in an effective way.

2. Students understood the concept which was reflected from their answers during sharing session.

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting the activity
2. Slow learners were not able to recollect and present some points during sharing session.

References:

- ✓ Allan Greenwood, 'Electrical Transients in Power Systems', Wiley Inter Science, New York, 2nd Edition, 1991.
- ✓ Pritindra Chowdhari, "Electromagnetic transients in Power System", John Wiley and Sons Inc.,Second Edition, 2009.
- ✓ https://www.readingrockets.org/strategies/think-pair-share

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Course Code & Title: EE3037 Power System Transients

Name of the Faculty member: Dr.D.Karthik Prabhu

Innovative Practice Description

- **Unit / Topic:** Unit III / Factors Contributing to good line design
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** One Minute Paper

- **Justification:**

The chosen topic – factors contributing to good line design. Since, in power system network design it should be considering some factors. After teaching the concept, I thought of conducting this activity for enabling the students to understand the transmission line design consideration which enhance their learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 10 Minutes

- **Details of the Implementation:**

After completing the topic, I gave 5 minutes for the students to think about the topic without writing anything.

Total Strength: 53

Reporter : Myself

At the end of the class

- ✓ I asked the students to think about factors contributing for good transmission line design 5 minutes.
- ✓ I told them to write as much as they remember for 3 minutes.
- ✓ Finally, I collected the papers from the students.(2 minutes)

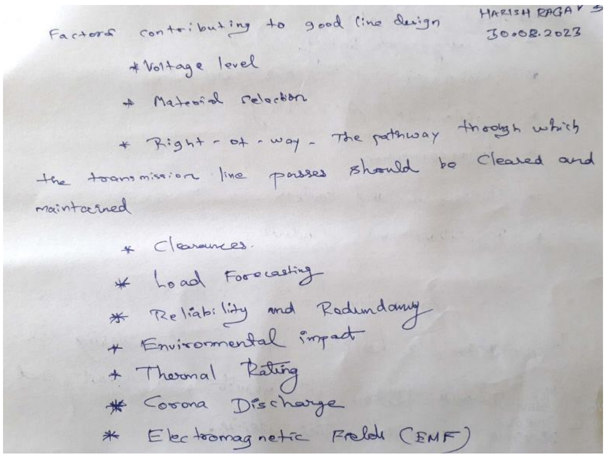
- **CO – PO / PSO mapping:**

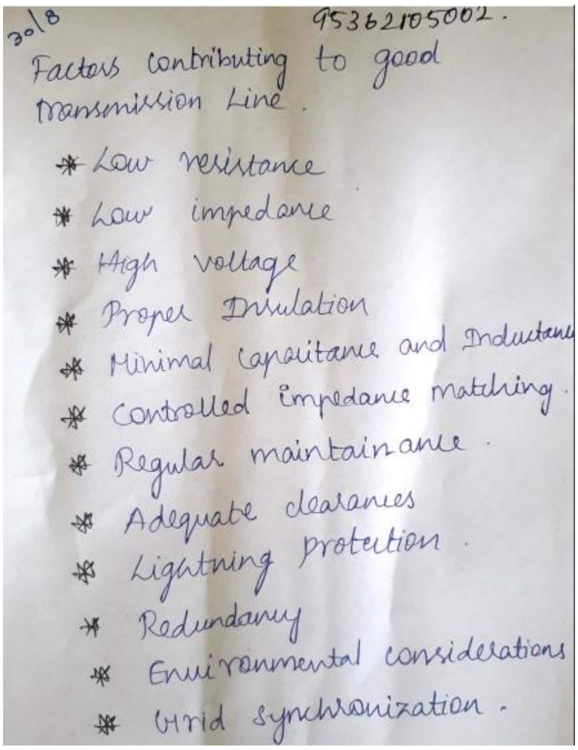
CO	PO1	PO2	PO3	PO5	PO9	PO10	PO12	PSO1
CO1	3	2	3	3	1	1	1	2

- **PO / PSO mapped:**

<table>
<tr><td rowspan="2">Innovative practice</td><td>PO10</td></tr>
<tr><td>1</td></tr>
<tr><td>Justification for correlation</td><td>Due to this activity, student's written communication will be enhanced. So it is slightly correlated.</td></tr>
</table>

- **Screenshot of the activity:**


HARISH RAGAV S
30.08.2023
Factors contributing to good line design
* Voltage level
* Material selection
* Right - of - way - The pathway through which the transmission line passes should be cleared and maintained
* Clearances.
* Load Forecasting
* Reliability and Redundancy
* Environmental impact
* Thermal Rating
* Corona Discharge
* Electromagnetic Fields (EMF)


2018
953621105002.
Factors contributing to good transmission Line.
* Low resistance
* Low impedance
* High voltage
* Proper Insulation
* Minimal capacitance and Inductance
* Controlled impedance matching.
* Regular maintainance.
* Adequate clearances
* Lightning protection.
* Redundancy
* Environmental considerations
* Grid synchronization.

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

Students felt that writing about the given topic make them understand it well and able to recollect immediately.

❖ ***Benefit of the practice:***

1. All the students were able to write the given topic.
2. After conducting the activity, I came to know that students understood the topic and they were able to explain it.
3. During IAT-II most of the students answered correctly in PART-A.

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting the activity
2. Some students could not be able to recollect and write immediately.

References:

✓ Allan Greenwood, 'Electrical Transients in Power Systems', Wiley Inter Science, New York, 2nd Edition, 1991.

✓ Pritindra Chowdhari, "Electromagnetic transients in Power System", John Wiley and Sons Inc.,Second Edition, 2009.

✓ https://oncourseworkshop.com/self-awareness/one-minute-paper/

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